

Nuclear Winter Data Commons Project Brief

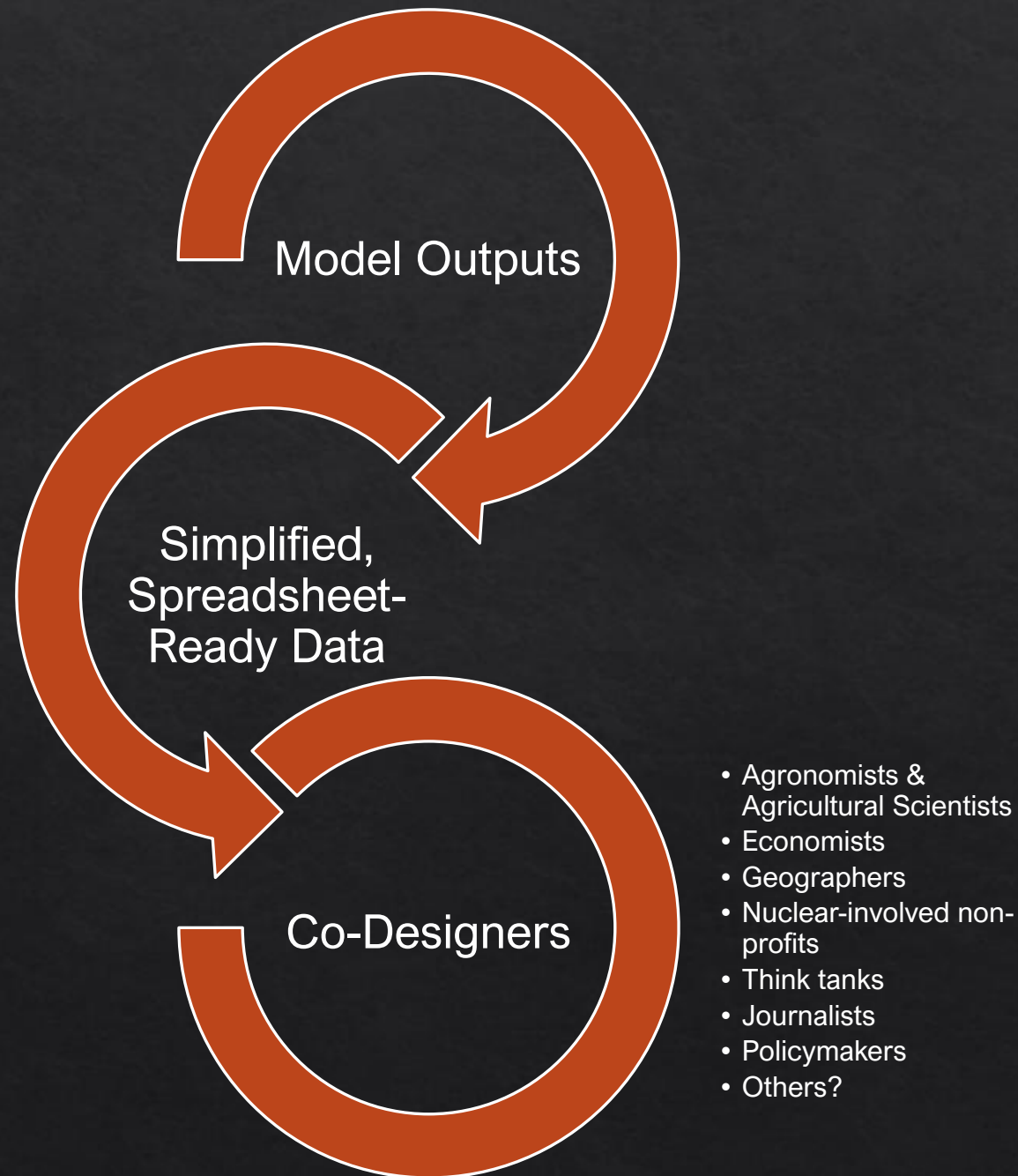
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Image source: Midjourney

Project Goal

Facilitate broader access to nuclear war climate simulation model outputs for a range of users. A co-design framework implies working with users to create & iteratively update relevant, useful data products.



Initial Products

1. Publication
2. Publicly available simplified tables (OSF)
3. Original model outputs

Potential Products

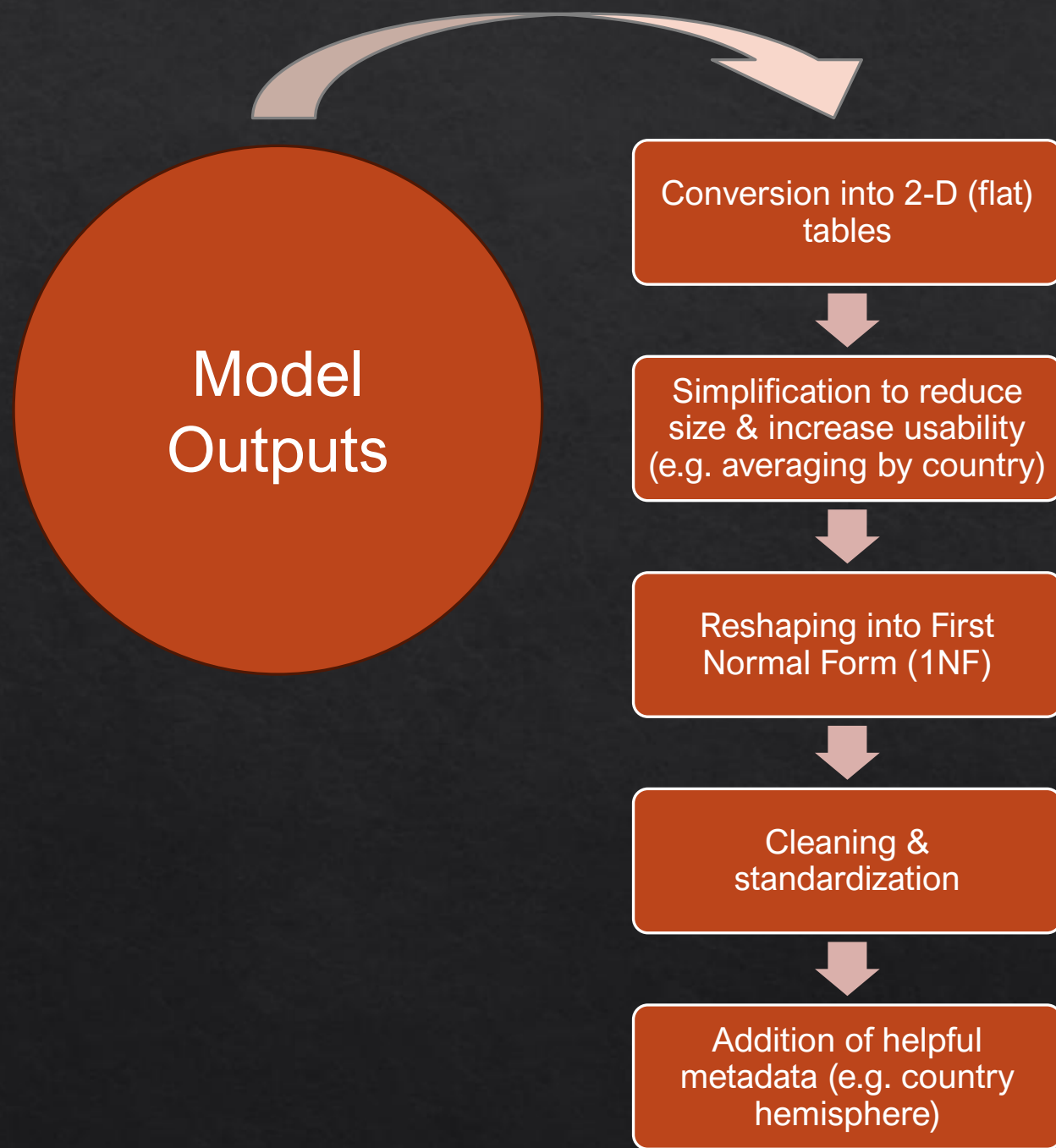
1. Dashboard(s)
2. API
3. Custom Datasets
4. Other ideas?

Current Data Structure - Themes

The first-generation repository covers seven themes. The first table covers the two most major climatic effects of stratospheric soot injection, changing temperature & precipitation, while the third covers the lesser-studied but quite important theme of increased ultraviolet radiation due to ozone depletion. Two tables contain data on changing agricultural yields and expected casualties from starvation during year 2 after the conflict. The next tables cover predictions of fish catch and predictions of sea ice thickness surrounding major world ports.

Theme:	Temperature, Precipitation	UV	Crop Yields (CLM)	Crop Yields (AGMIP)	Starvation	Fish Catch	Sea Ice
Geographic Unit:	Country	Country	Country	Country	Country	EEZ (exclusive economic zone)	Major Port
Time Unit:	Month	Month	Year	Year	Year 2 (only)	Year	Month

Current Data Processing

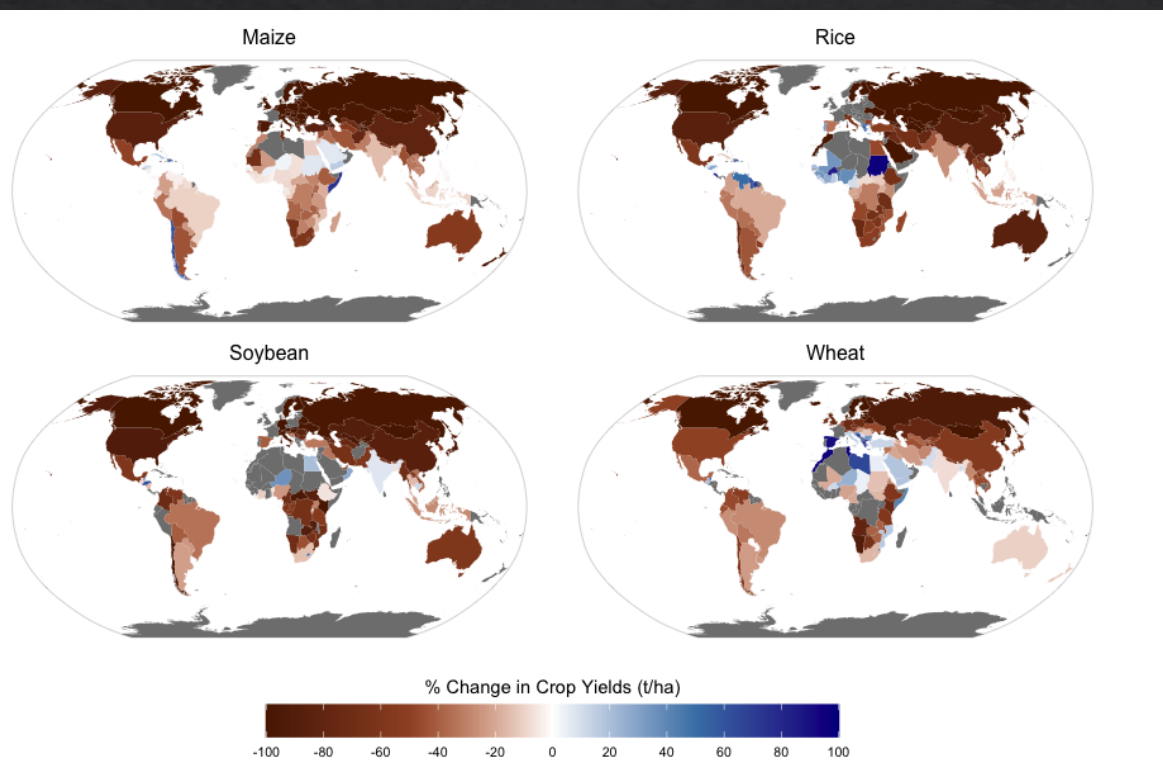


Final Tables Available through OSF

Country	Month	Indicator	Value
Afghanistan	1	UVA	10.01
Albania	1	UVA	6.17
Algeria	1	UVA	12.83
American Samoa	1	UVA	NA
Antigua & Barbuda	1	UVA	21.58
Afghanistan	2	UVA	12.85
Albania	2	UVA	8.83
Algeria	2	UVA	15.76
American Samoa	2	UVA	NA
Antigua & Barbuda	2	UVA	21.25
Afghanistan	1	UVB	0.22
Albania	1	UVB	0.09
Algeria	1	UVB	0.35
...	...	...	...
Antigua & Barbuda	2	UVB	0.95

Example Analyses

Percent change of crop production for four major crops (maize, rice, soybean, and wheat) by country level for the 37 Tg scenario



Sea Ice Thickness in Shanghai Harbor
(No sea ice under normal conditions)



Table	Theme	Geographic Unit (a)	Time Unit (b)	Soot Injection Scenario (c)	Associated Publication: Earth System Simulation Reference (d)	Association Publication: Analysis & Discussion (e)
1. Temperature & Precipitation	Temperature	Grid cell (lat, long)	Month	150 Tg	2. Coupe et al., 2019 (f)	2. Coupe et al., 2019
				16 Tg	7. Toon et al., 2019	7. Toon et al., 2019
				27 Tg	7. Toon et al., 2019	7. Toon et al., 2019
				37 Tg	7. Toon et al., 2019	7. Toon et al., 2019
				47 Tg	7. Toon et al., 2019	7. Toon et al., 2019
				5 Tg	7. Toon et al., 2019	7. Toon et al., 2019
				control	7. Toon et al., 2019	7. Toon et al., 2019
	Precipitation	Grid cell (lat, long)	Month	150 Tg	2. Coupe et al., 2019	2. Coupe et al., 2019
				16 Tg	7. Toon et al., 2019	7. Toon et al., 2019
				27 Tg	7. Toon et al., 2019	7. Toon et al., 2019
				37 Tg	7. Toon et al., 2019	7. Toon et al., 2019
				47 Tg	7. Toon et al., 2019	7. Toon et al., 2019
				5 Tg	7. Toon et al., 2019	7. Toon et al., 2019
				control	7. Toon et al., 2019	7. Toon et al., 2019
3. UV	UV	Country	Month	150 Tg	2. Coupe et al., 2019	1. Bardeen et al., 2021
				control	2. Coupe et al., 2019	1. Bardeen et al., 2021
4a. Agriculture (CLM)	Agriculture	Country	Year	150 Tg	2. Coupe et al., 2019	8. Xia et al., 2015
				16 Tg	7. Toon et al., 2019	8. Xia et al., 2015
				27 Tg	7. Toon et al., 2019	8. Xia et al., 2015
				37 Tg	7. Toon et al., 2019	8. Xia et al., 2015
				47 Tg	7. Toon et al., 2019	8. Xia et al., 2015
				5 Tg	7. Toon et al., 2019	8. Xia et al., 2015
4b. Agriculture (AGMIP)	Agriculture	Country	Year	5 Tg	5. Mills et al., 2014	4. Jägermeyr et al., 2020
				5 Tg	7. Toon et al., 2019	4. Jägermeyr et al., 2020
5. Fish Catch	Fish Catch	EEZ (exclusive economic zone)	Year	150 Tg	2. Coupe et al., 2019	6. Scherrer et al., 2020
				16 Tg	7. Toon et al., 2019	6. Scherrer et al., 2020
				27 Tg	7. Toon et al., 2019	6. Scherrer et al., 2020
				37 Tg	7. Toon et al., 2019	6. Scherrer et al., 2020
				47 Tg	7. Toon et al., 2019	6. Scherrer et al., 2020
				5 Tg	7. Toon et al., 2019	6. Scherrer et al., 2020
				control	7. Toon et al., 2019	6. Scherrer et al., 2020
6. Sea Ice	Sea Ice	Port	Month	150 Tg	2. Coupe et al., 2019	3. Harrison et al., 2022
				37 Tg	7. Toon et al., 2019	3. Harrison et al., 2022
				47 Tg	7. Toon et al., 2019	3. Harrison et al., 2022

(a) Smallest (most disaggregated) geographic unit in the dataset.

(b) Smallest (most disaggregated) time unit in the dataset.

(c) Mass of black carbon soot injected into the stratosphere in teragrams (1Tg = 10¹²g = 1 trillion grams).

(d) The publication for which the earth system simulation was originally run. These earth systems simulations use sub-model components for atmosphere, aerosols, land, and ocean responses. For papers on temperature and precipitation, this column will be the same as the final column. For all other themes, additional sub-model components (e.g. for crops) were run using the outputs from the original simulations and the results discussed in separate papers (see next footnote).

(e) The publication that analyzes and discusses results of the complete earth systems model, including all relevant sub-model components.

(f) Publication citations are abbreviated in the table to save space. Full citations:

1. Bardeen, Charles G., Douglas E. Kinnison, Owen B. Toon, Michael J. Mills, Francis Vitt, Lili Xia, Jonas Jägermeyr, et al. “Extreme Ozone Loss Following Nuclear War Results in Enhanced Surface Ultraviolet Radiation.” *Journal of Geophysical Research: Atmospheres* 126, no. 18 (September 27, 2021): e2021JD035079. <https://doi.org/10.1029/2021JD035079>.

2. Coupe, Joshua, Charles G. Bardeen, Alan Robock, and Owen B. Toon. “Nuclear Winter Responses to Nuclear War Between the United States and Russia in the Whole Atmosphere Community Climate Model Version 4 and the Goddard Institute for Space Studies ModelE.” *Journal of Geophysical Research: Atmospheres* 124, no. 15 (August 16, 2019): 8522–43. <https://doi.org/10.1029/2019JD030509>.

3. Harrison, Cheryl S., Tyler Rohr, Alice DuVivier, Elizabeth A. Maroon, Scott Bachman, Charles G. Bardeen, Joshua Coupe, et al. “A New Ocean State After Nuclear War.” *AGU Advances* 3, no. 4 (August 2022): e2021AV000610. <https://doi.org/10.1029/2021AV000610>.

4. Jägermeyr, Jonas, Alan Robock, Joshua Elliott, Christoph Müller, Lili Xia, Nikolay Khabarov, Christian Folberth, et al. “A Regional Nuclear Conflict Would Compromise Global Food Security.” *Proceedings of the National Academy of Sciences* 117, no. 13 (March 31, 2020): 7071–81. <https://doi.org/10.1073/pnas.1919049117>.

5. Mills, Michael J., Owen B. Toon, Julia Lee-Taylor, and Alan Robock. “Multidecadal Global Cooling and Unprecedented Ozone Loss Following a Regional Nuclear Conflict.” *Earth’s Future* 2, no. 4 (April 2014): 161–76. <https://doi.org/10.1002/2013EF000205>.

6. Scherrer, Kim J. N., Cheryl S. Harrison, Ryan F. Heneghan, Eric Galbraith, Charles G. Bardeen, Joshua Coupe, Jonas Jägermeyr, et al. “Marine Wild-Capture Fisheries after Nuclear War.” *Proceedings of the National Academy of Sciences* 117, no. 47 (November 24, 2020): 29748–58. <https://doi.org/10.1073/pnas.2008256117>.

7. Toon, Owen B., Charles G. Bardeen, Alan Robock, Lili Xia, Hans Kristensen, Matthew McKinzie, R. J. Peterson, Cheryl S. Harrison, Nicole S. Lovenduski, and Richard P. Turco. “Rapidly Expanding Nuclear Arsenals in Pakistan and India Portend Regional and Global Catastrophe.” *Science Advances* 5, no. 10 (October 11, 2019): eaay5478. <https://doi.org/10.1126/sciadv.aay5478>.

8. Xia, Lili, Alan Robock, Michael Mills, Andrea Stenke, and Ira Helfand. “Decadal Reduction of Chinese Agriculture after a Regional Nuclear War.” *Earth’s Future* 3, no. 2 (February 2015): 37–48. <https://doi.org/10.1002/2014EF000283>.